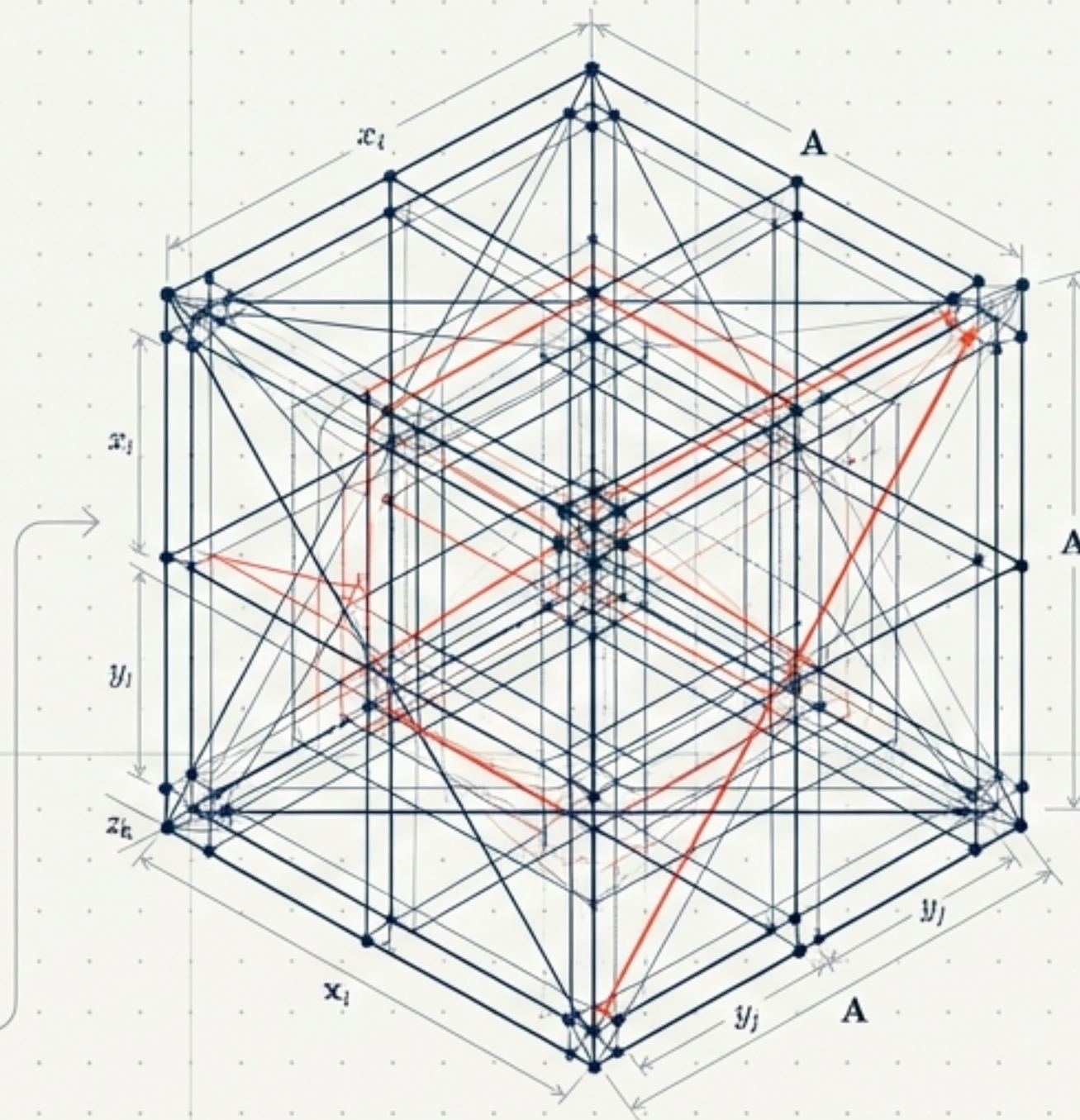
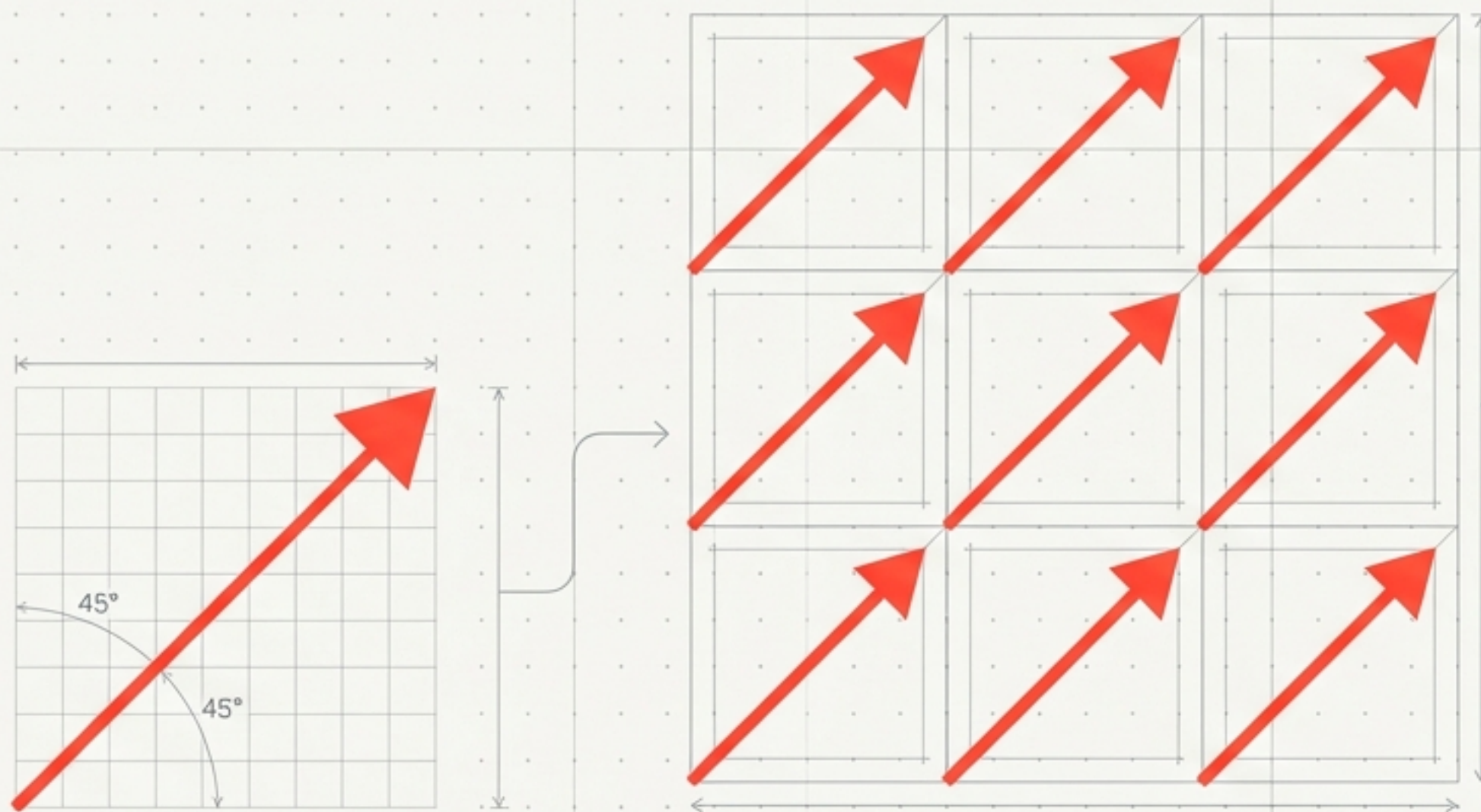


From Arrows to Algorithms: The Language of Data

Understanding the mathematical engine behind
Data Science, Statistics, and Machine Learning



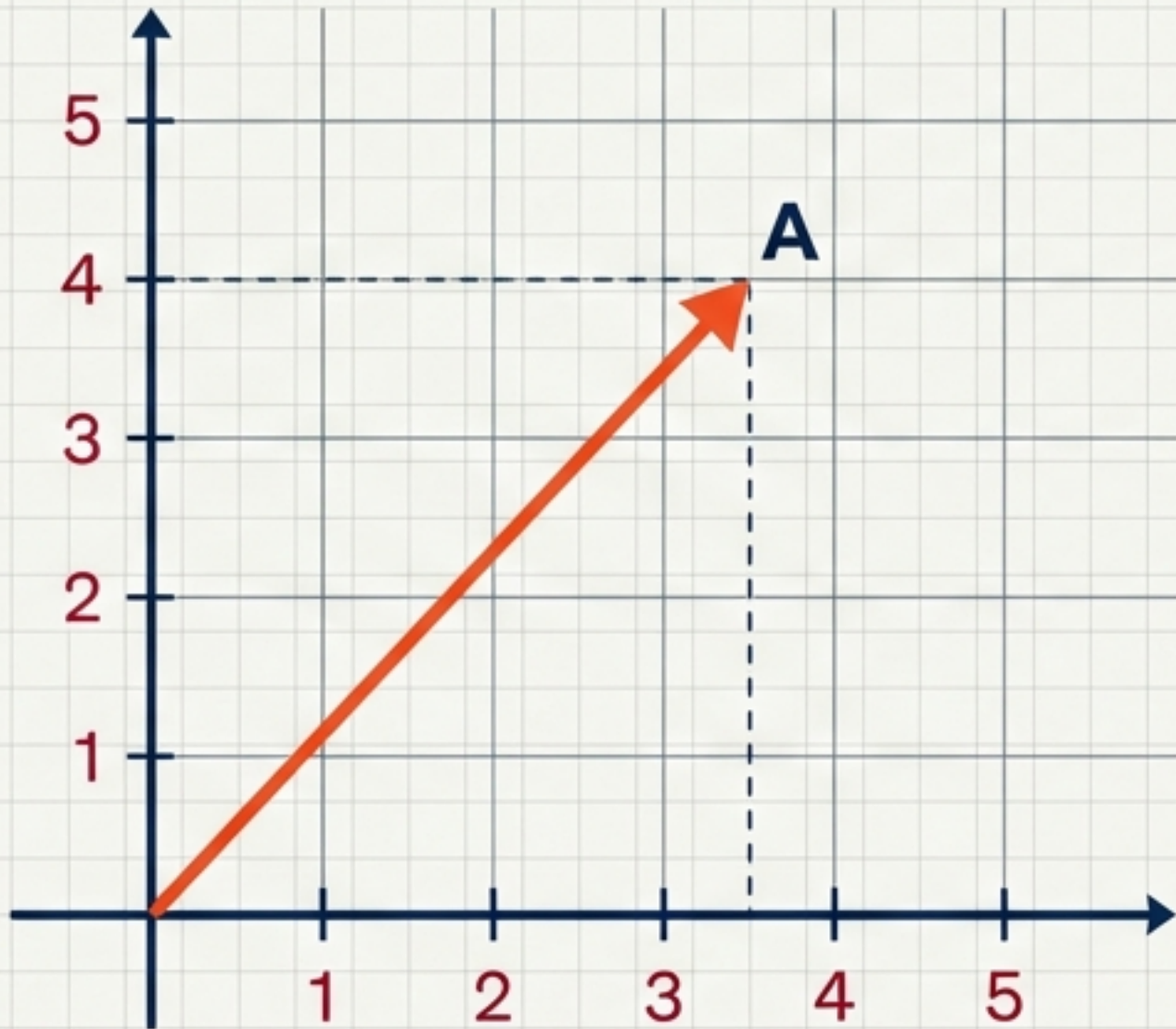
$y = Wx + b$

Linear Algebra is the foundational infrastructure for everything from predicting house prices to training neural networks. While standard Algebra deals with single numbers (scalars), Linear Algebra manages equations of vectors and matrices—data in multiple dimensions.

$y = Wx + b$

The Atom of Data: The Vector

Geometric Representation: An arrow in space.



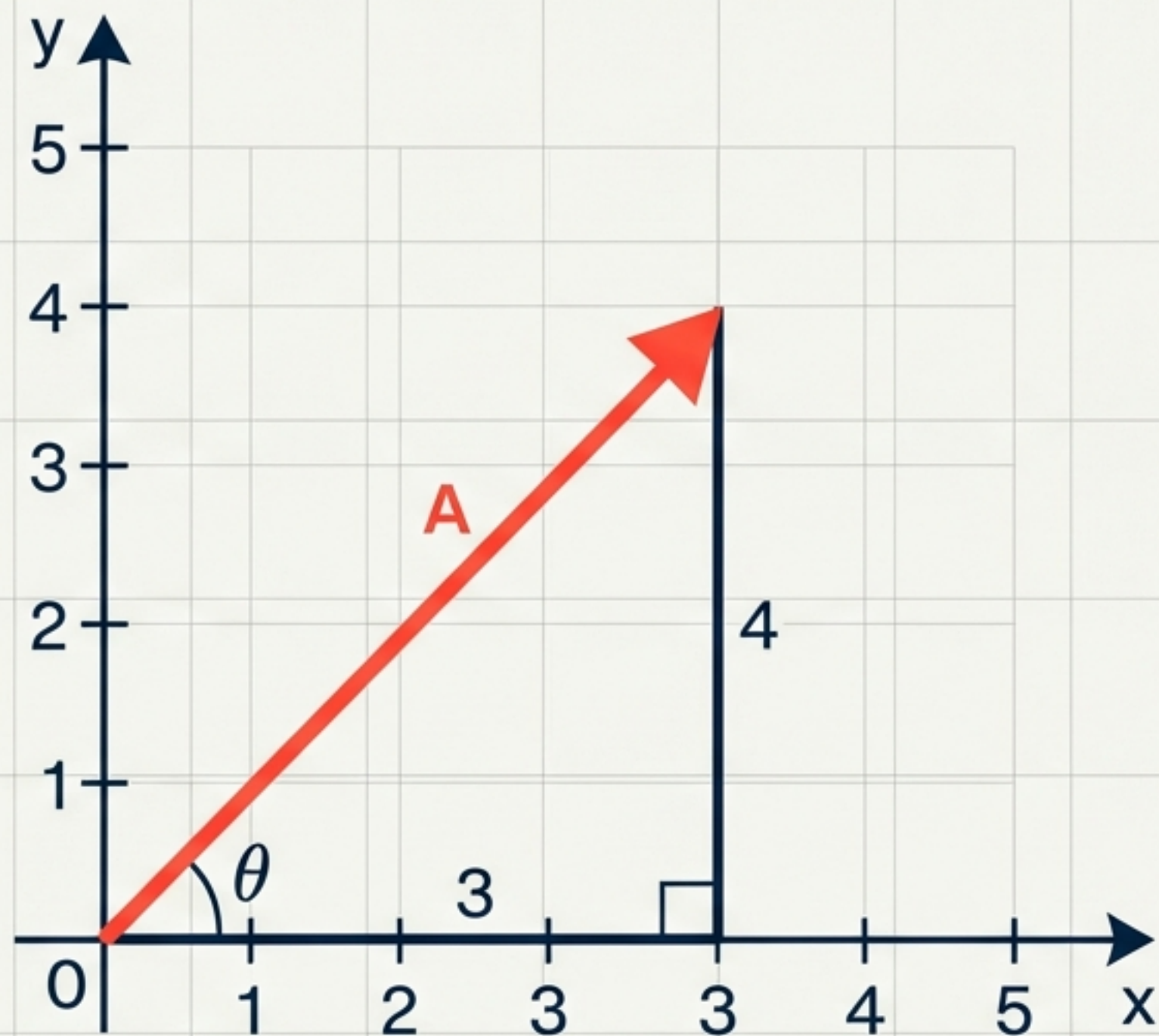
Algebraic View: A list of numbers.

$$\text{Vector } \mathbf{A} = (3, 4)$$

- A quantity possessing both:
 1. Magnitude (Length)
 2. Direction (Angle)

Key Insight: This duality allows us to visualise complex data problems as spatial geometry.

Anatomy of a Vector



Worked Example

Property 1: Magnitude (Length)

Derived from the Pythagorean theorem:

$$|\mathbf{A}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

Property 2: Direction (Angle)

The angle relative to the x-axis:

$$\theta = \tan^{-1}(4/3) \approx 53.13^\circ$$

Data Shapes: Row vs. Column Vectors

The Row Vector (Observation)



$[1.80\text{m}, 75\text{kg}, 32\text{yrs}, \text{£}45\text{k}]$

Height

Weight

Age

Income

Structure: $1 \times n$ matrix.
Represents a single Observation
(e.g., one user profile).

1×4 row vector

The Column Vector (Variable)

$\begin{bmatrix} 4 \\ -1 \\ 6 \\ 2 \end{bmatrix} \leftarrow \mathbf{b}$

Structure: $n \times 1$ matrix.
Represents a Variable or
Feature across a dataset.

4×1 column vector

Operation I: Vector Addition

The Logic

Add corresponding components.

Input

Vector A = (3, 2) and Vector B = (1, 4)

Calculation

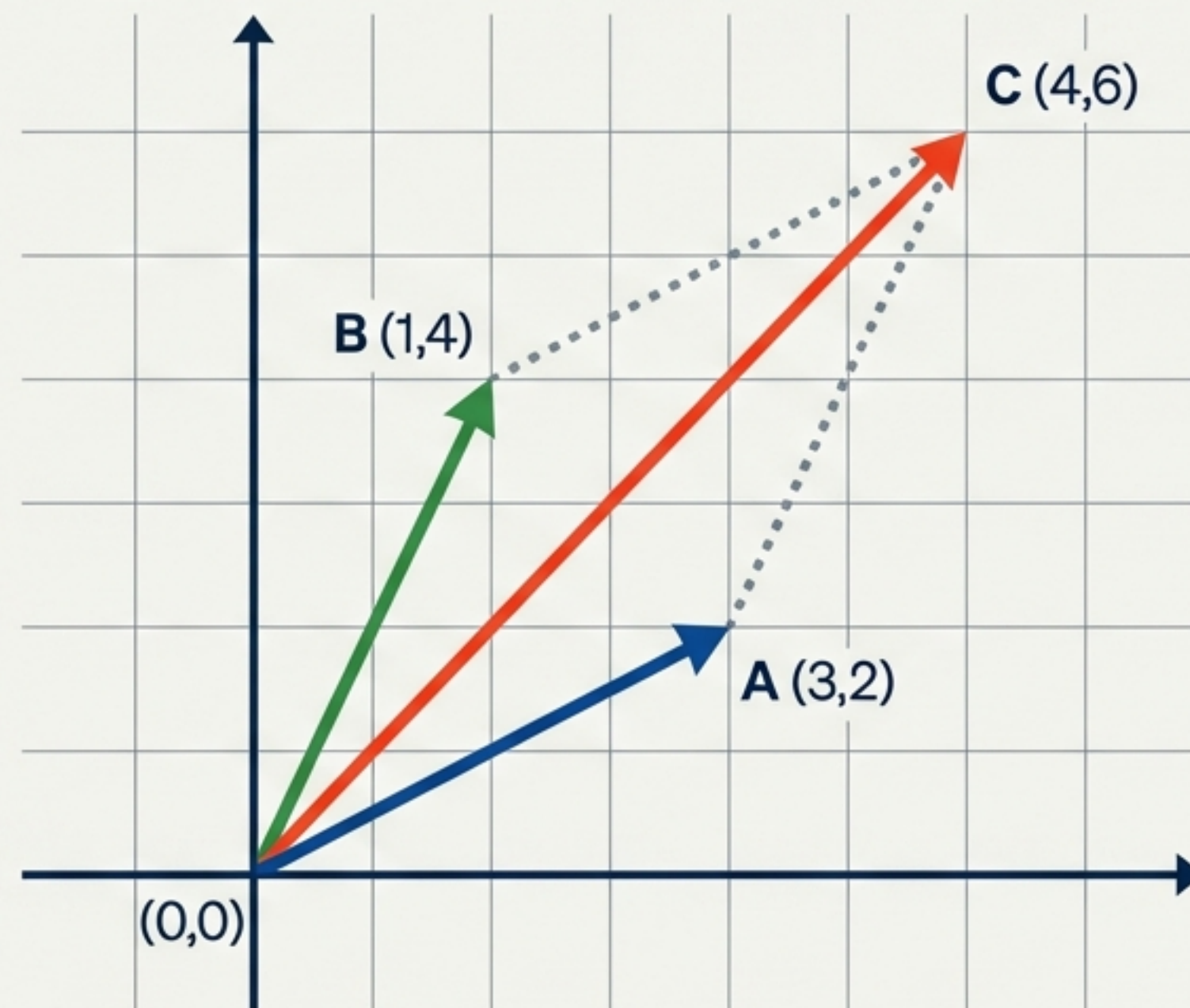
$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$\mathbf{C} = (3+1, 2+4) = (4, 6)$$

Resultant Magnitude

$$|\mathbf{C}| = \sqrt{52} \approx 7.21$$

The Visual



Operation II: Vector Subtraction

The Math

The Logic: Calculate displacement or difference.

Input: Vector A = (3, 2) and Vector B = (1, 4)

Calculation:

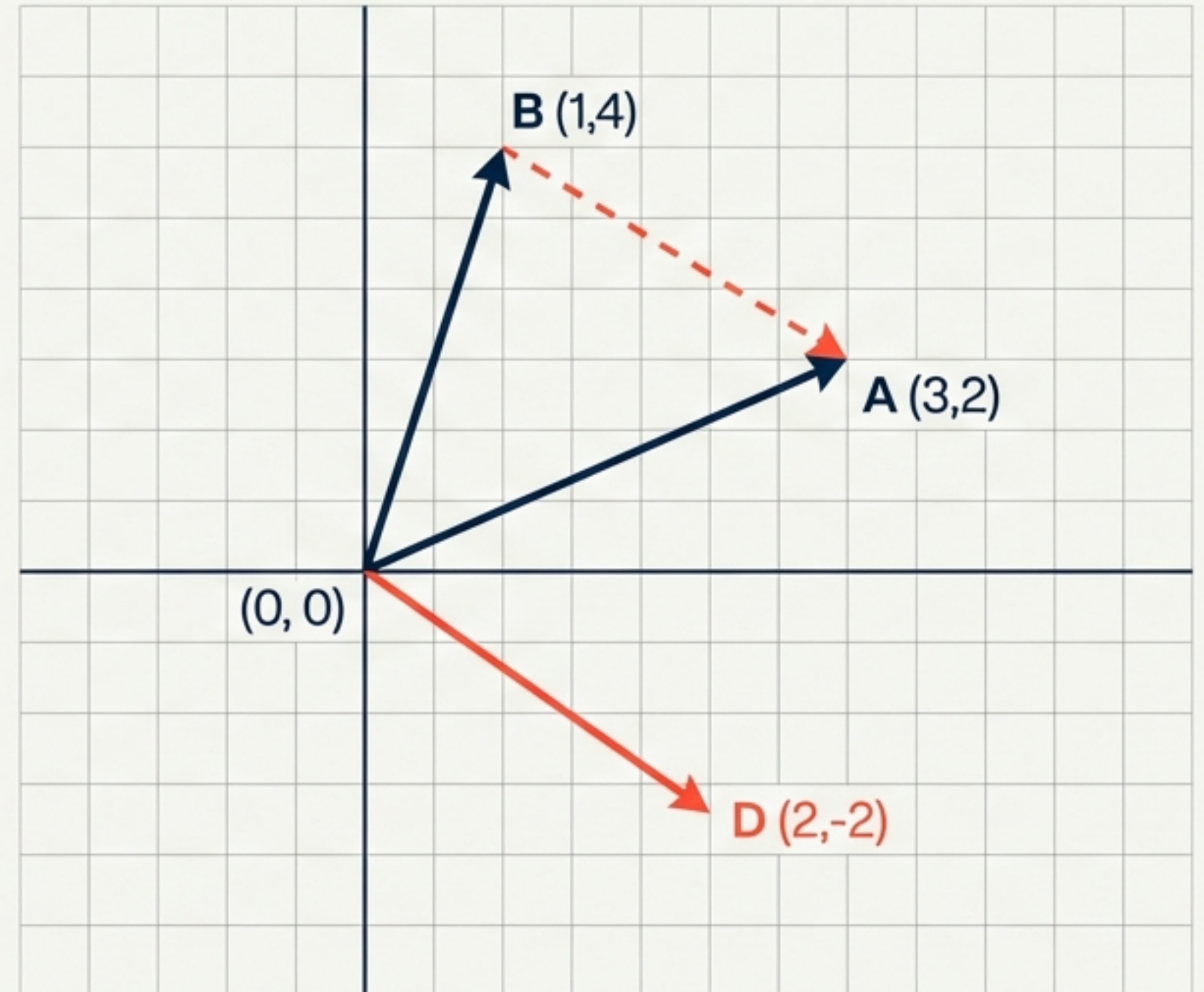
$$\mathbf{D} = \mathbf{A} - \mathbf{B}$$

$$\mathbf{D} = (3 - 1, 2 - 4) = (2, -2)$$

Resultant Magnitude:

$$|\mathbf{D}| = \sqrt{8} \approx 2.83$$

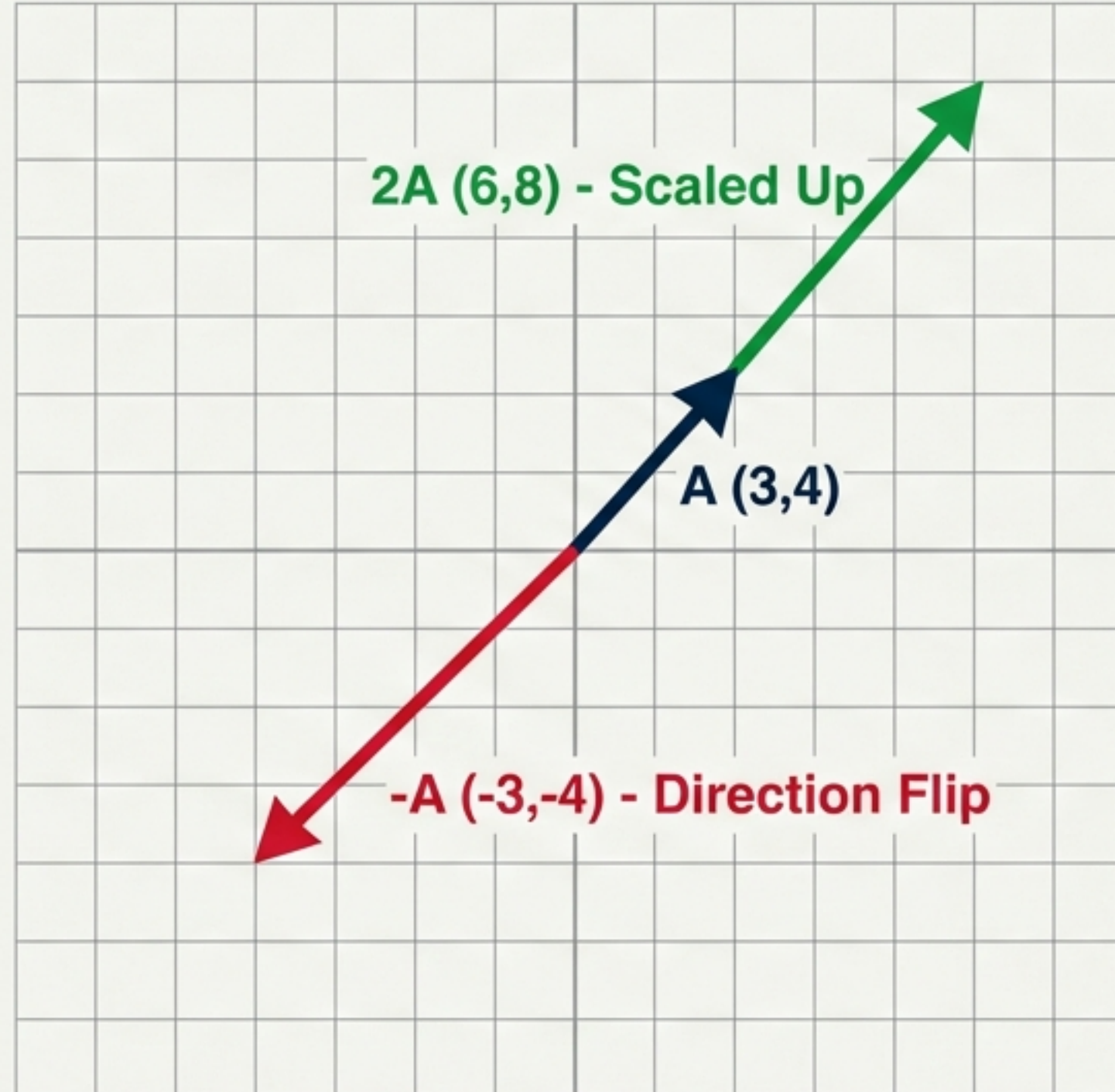
The Visual



Operation III: Scalar Operations

Scenario A: Scaling Up

Multiplying by 2 doubles the magnitude (length = 10) but preserves the direction.

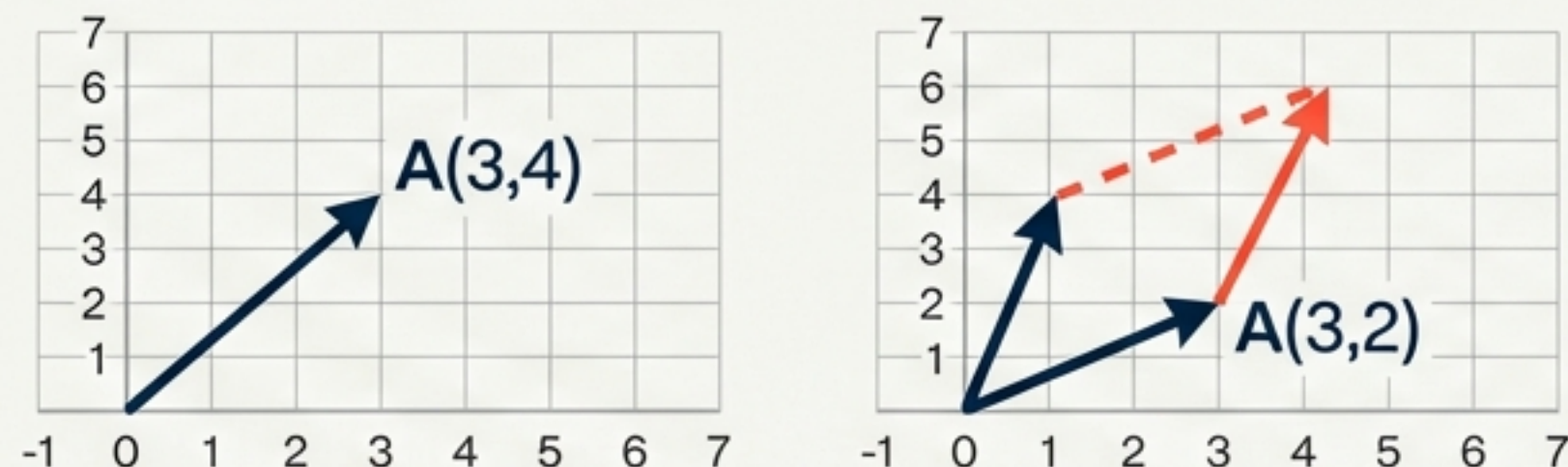


Scenario B: Direction Flip

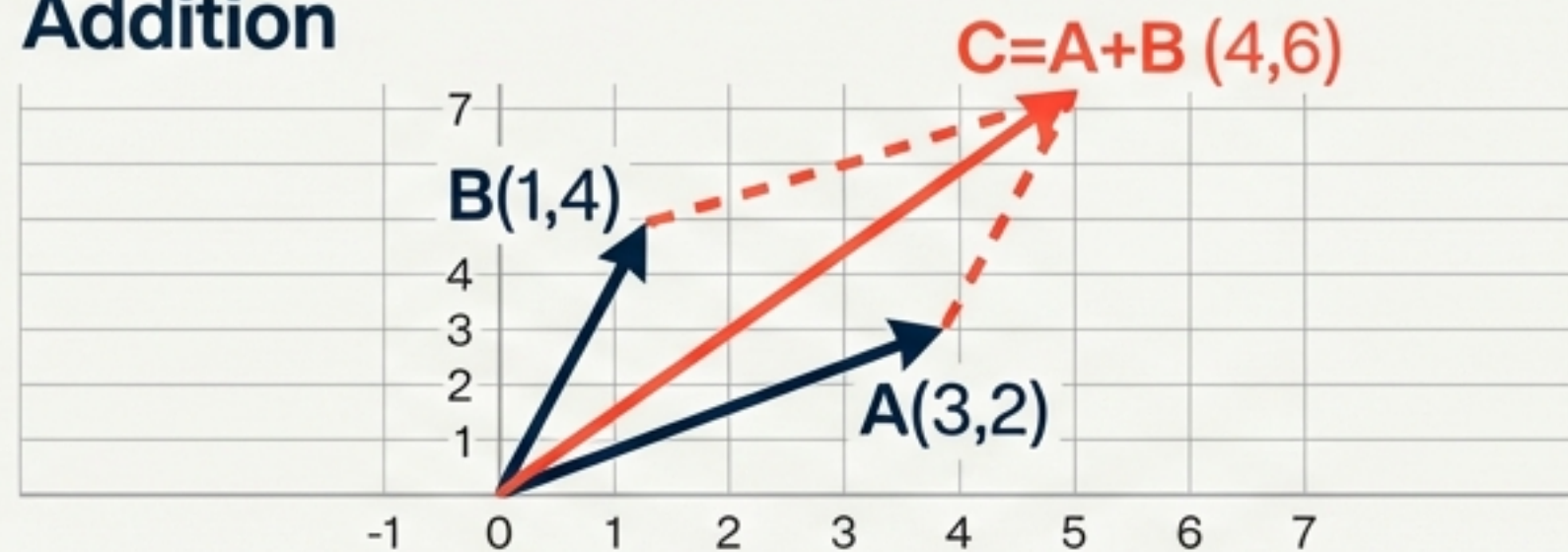
Multiplying by -1 preserves the length (5) but rotates the vector 180°.

The Geometric Sandbox

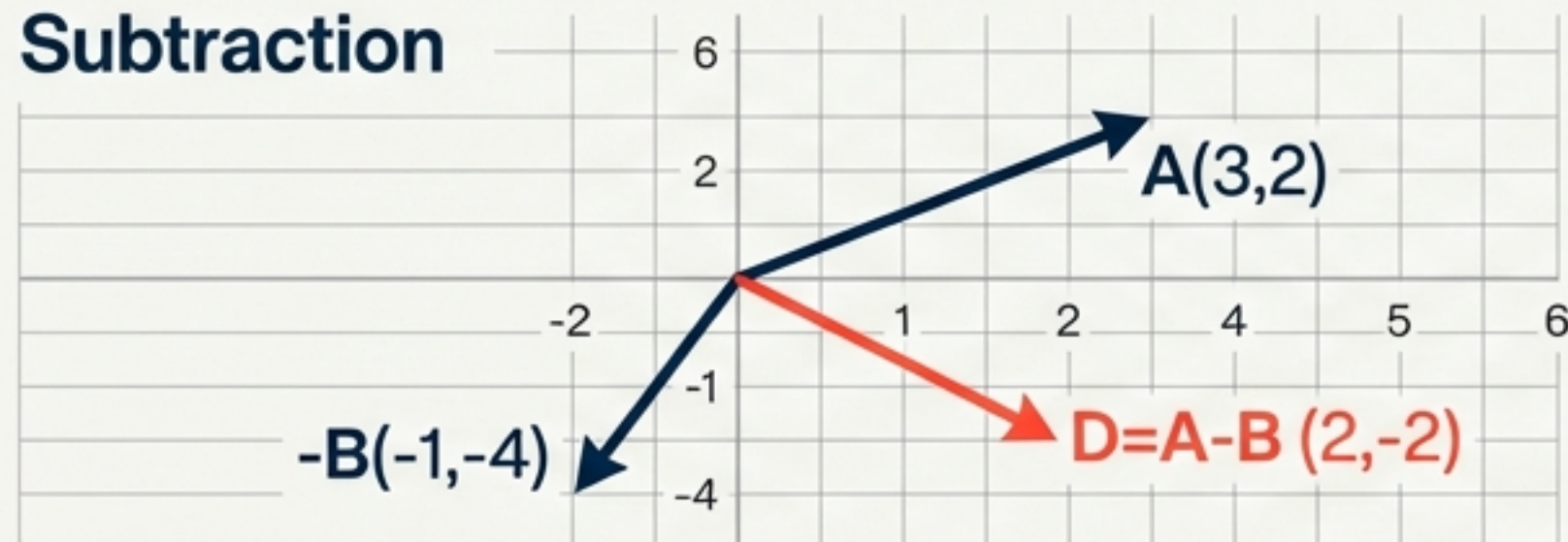
Representation



Addition



Subtraction

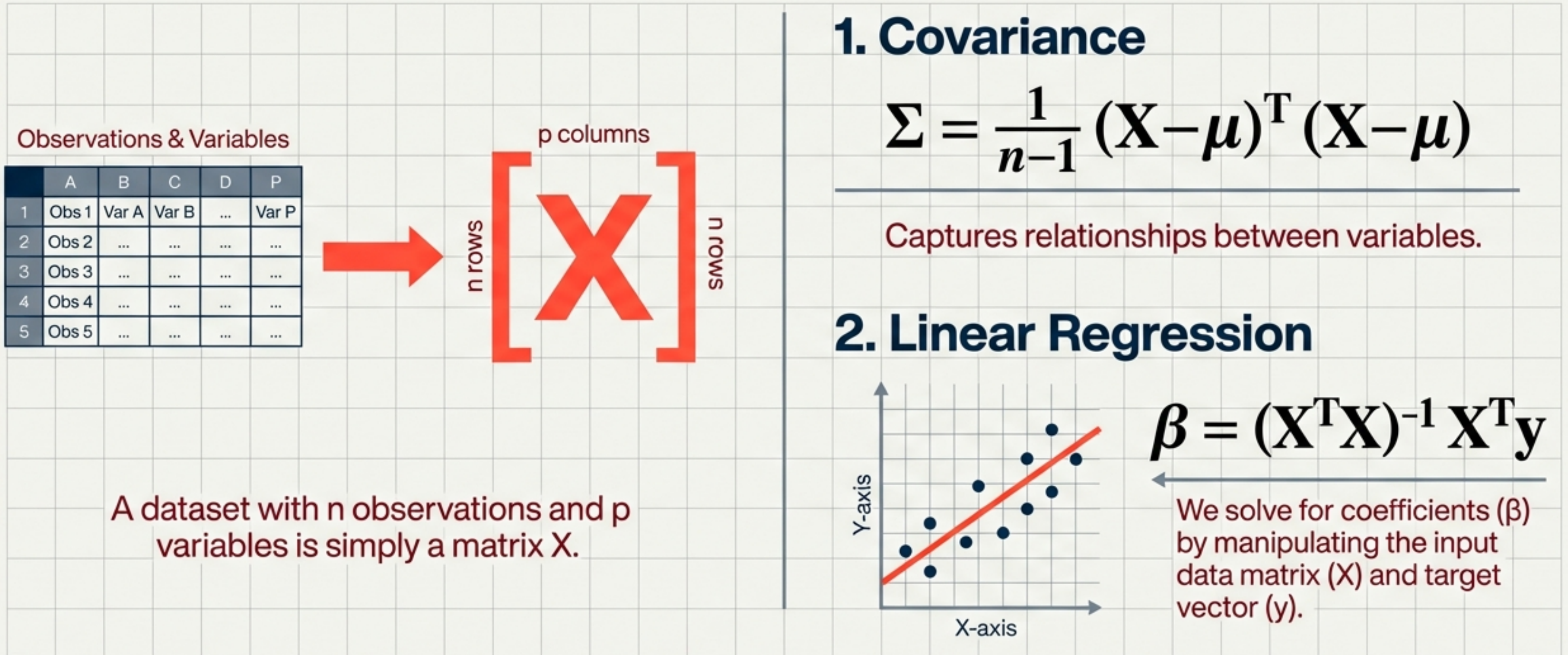


Scalar Ops



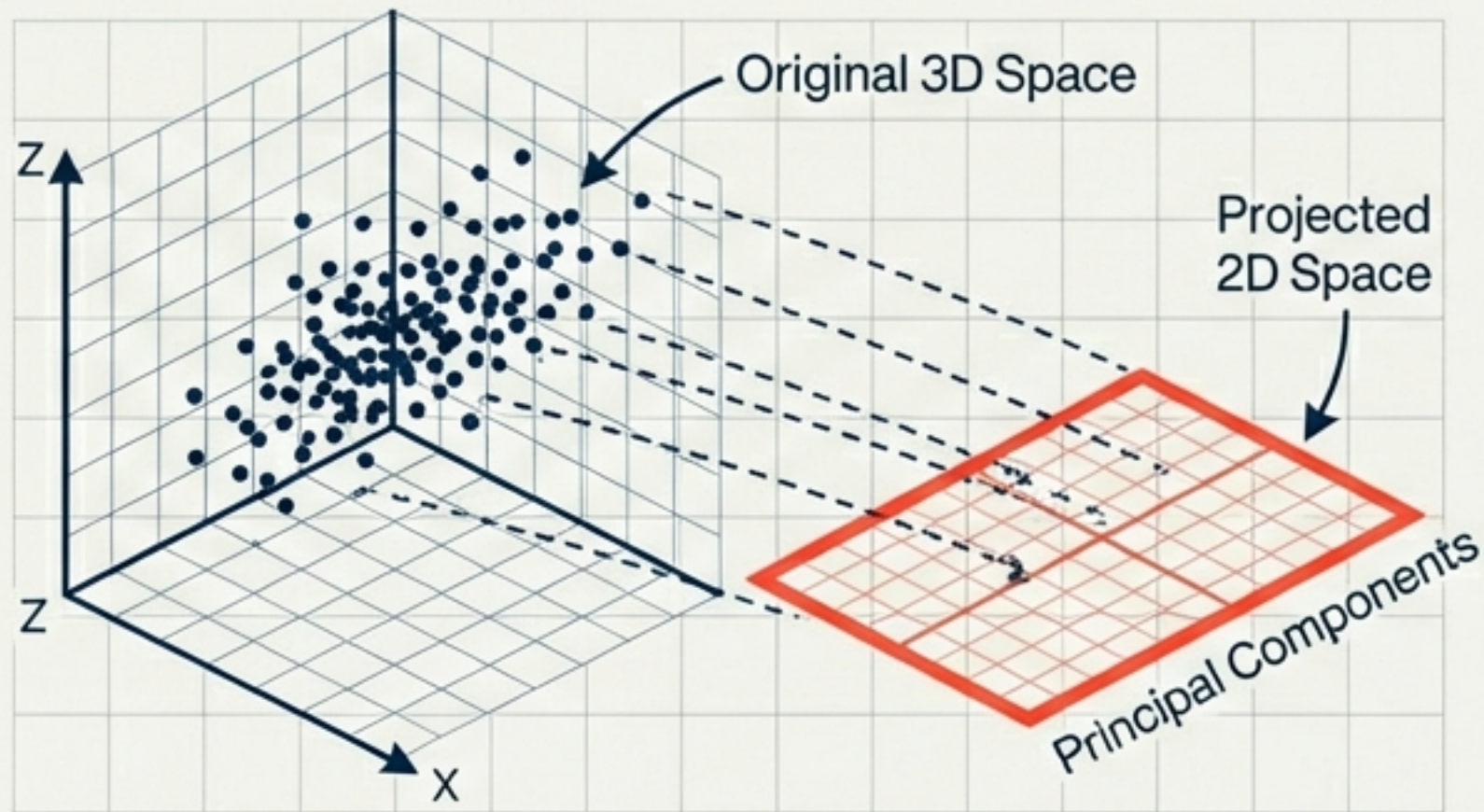
Takeaway: These simple geometric manipulations—adding, subtracting, stretching, and flipping—are the micro-steps that build complex algorithms.

Application: The Backbone of Statistics



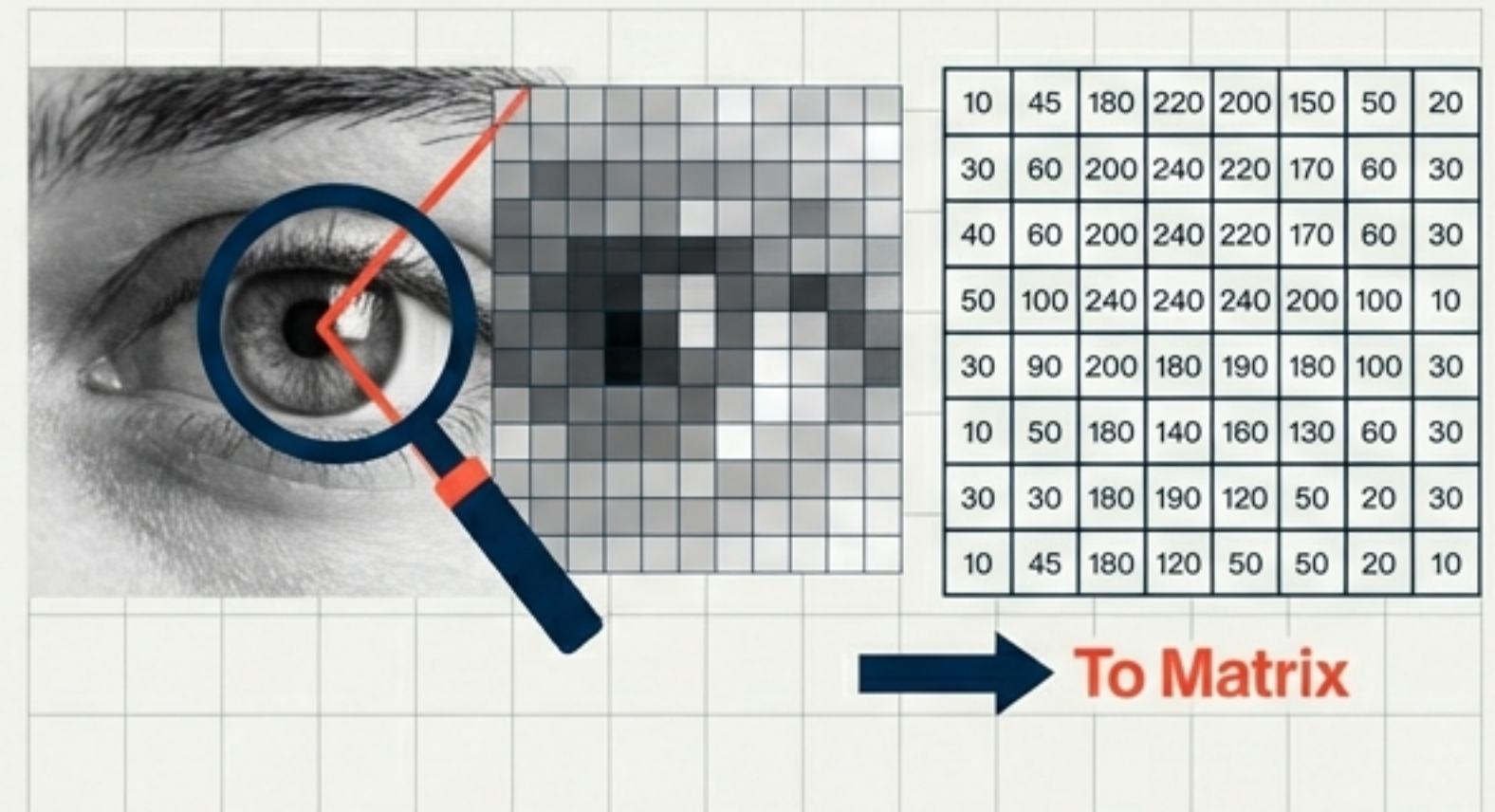
Application: Data Science & Transformation

Dimensionality Reduction (PCA)



Problem: Too many features (e.g., 50 customer attributes).
Solution: PCA uses Eigenvalues of the covariance matrix to compress data into principal components.

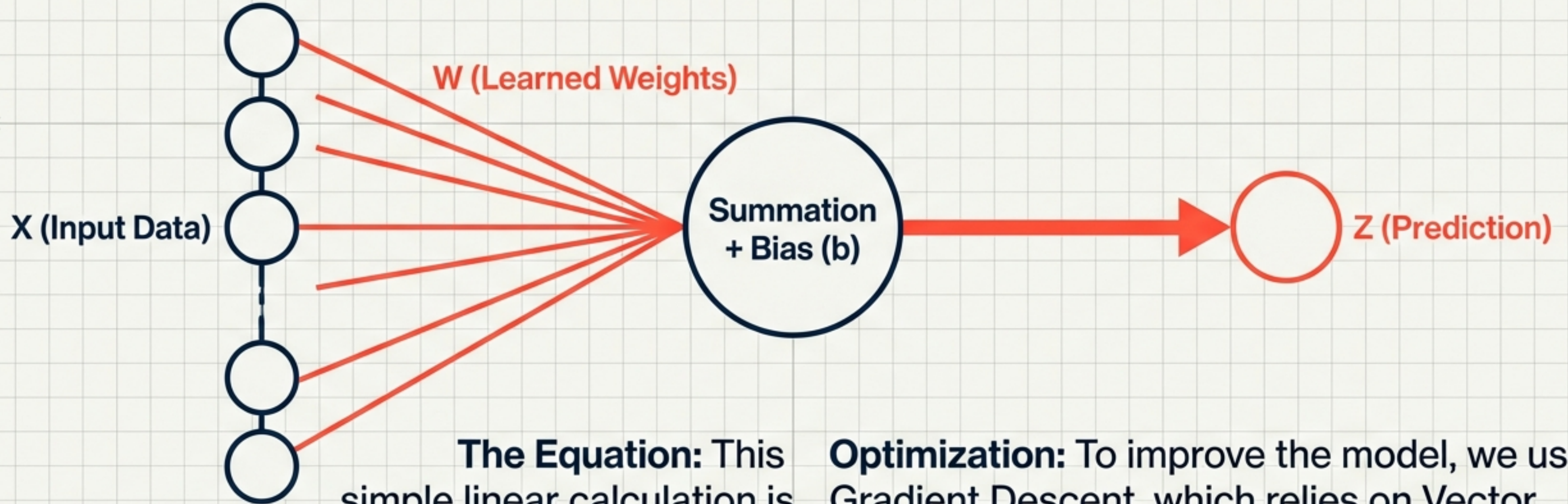
Image Representation



To a computer, an image is a matrix of pixel values.
Sharpening or edge detection is simply performing math on that matrix.

Application: Machine Learning & Neural Networks

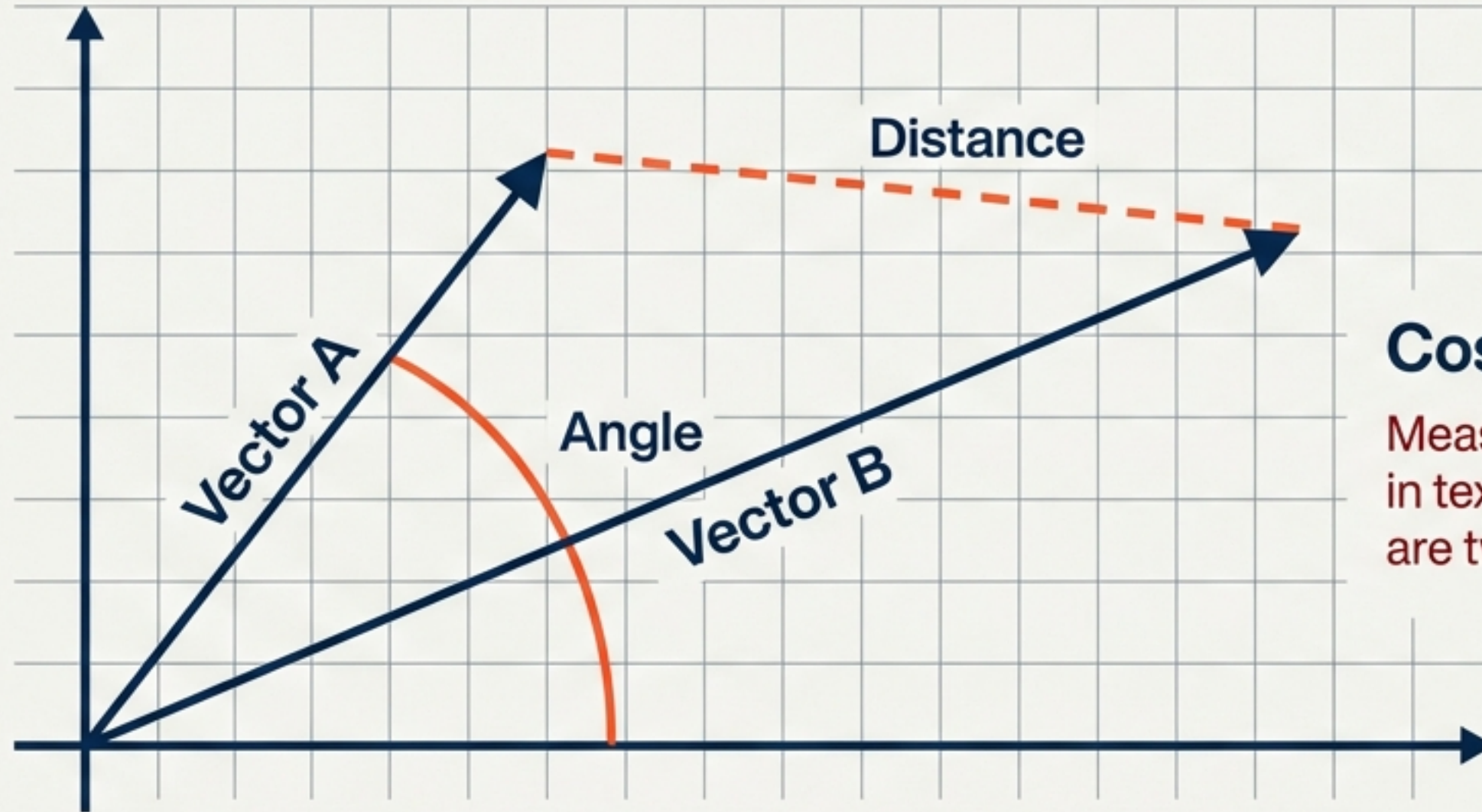
$$Z = W \cdot X + b$$



The Equation: This simple linear calculation is the core of deep learning.

Optimization: To improve the model, we use Gradient Descent, which relies on Vector Calculus to update 'W' and minimize error.

Application: Measuring Similarity



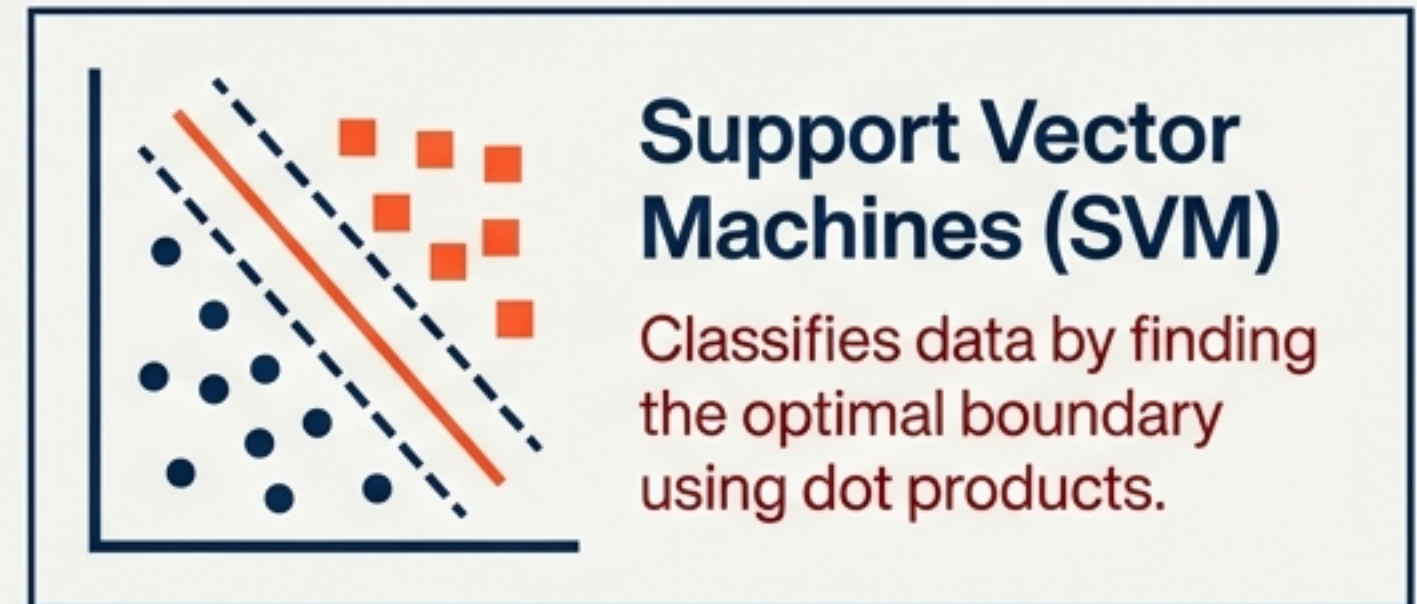
Cosine Similarity

Measures the angle. Used often in text analysis (e.g., how similar are two documents?).

Euclidean Distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

The straight-line physical distance between points.



Support Vector Machines (SVM)

Classifies data by finding the optimal boundary using dot products.

Practical Syntax: Python (NumPy)

```
import numpy as np

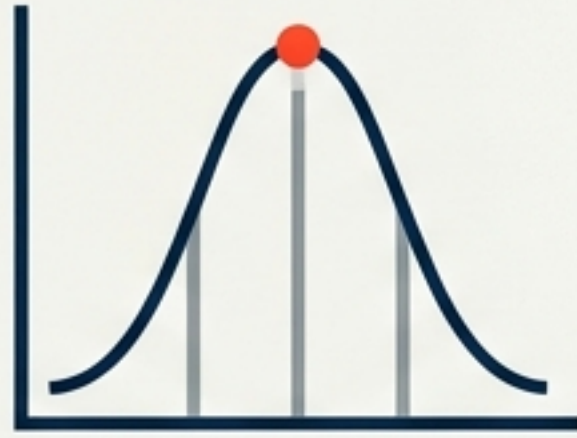
# Matrix of features (3 observations, 2 variables)
X = np.array([[1, 2], [3, 4], [5, 6]])

# Matrix Multiplication with Weights
W = np.array([[0.1], [0.2]])
y_pred = X @ W # Computes prediction
```

Vectors and matrices are treated as native arrays.

Performs the $Z = W \cdot X$ operation efficiently.

The Universal Language of Data



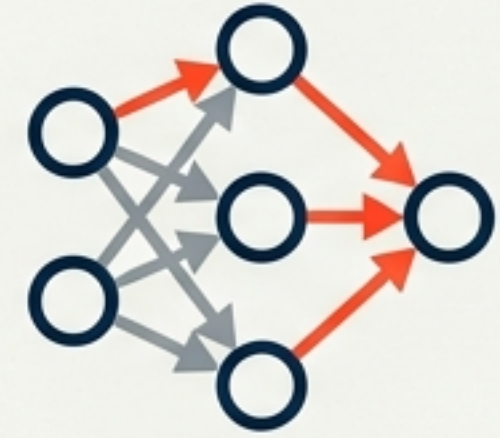
Statistics

Regression, Covariance,
Correlation.



Data Science

Dimensionality Reduction
(PCA), Image Processing.



Machine Learning

Neural Networks, Optimization,
Backpropagation.

When we view data not just as numbers, but as vectors and matrices, we unlock the ability to transform, model, and understand the world in multiple dimensions.